# FYP-LOGISTICS / PLC_1 [CPU 1214C DC/DC/Rly] / Program blocks

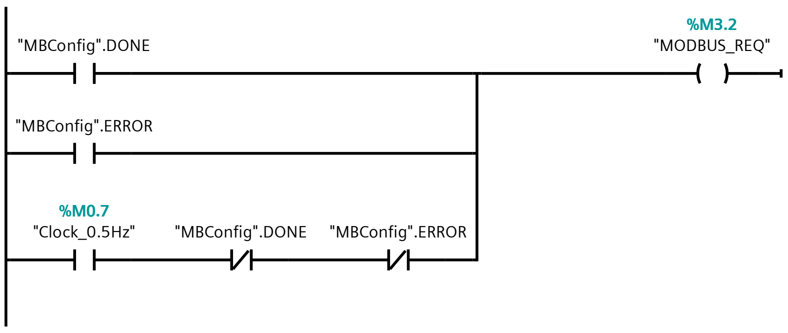
## Main [OB1]

| Main Properties | | | | | | | |
|---|---|---|---|---|---|---|---|
| **General** | | | | | | | |
| **Name** | Main | **Number** | 1 | **Type** | OB | **Language** | LAD |
| **Numbering** | Automatic | | | | | | |
| **Information** | | | | | | | |
| **Title** | "Main Program Sweep (Cycle)" | **Author** | | **Comment** | | **Family** | |
| **Version** | 0.1 | **User-defined ID** | | | | | |

| Name | Data type | Default value | Comment |
|---|---|---|---|
| Input | | | |
| Initial_Call | Bool | | Initial call of this OB |
| Remanence | Bool | | =True, if remanent data are available |
| Temp | | | |
| Constant | | | |

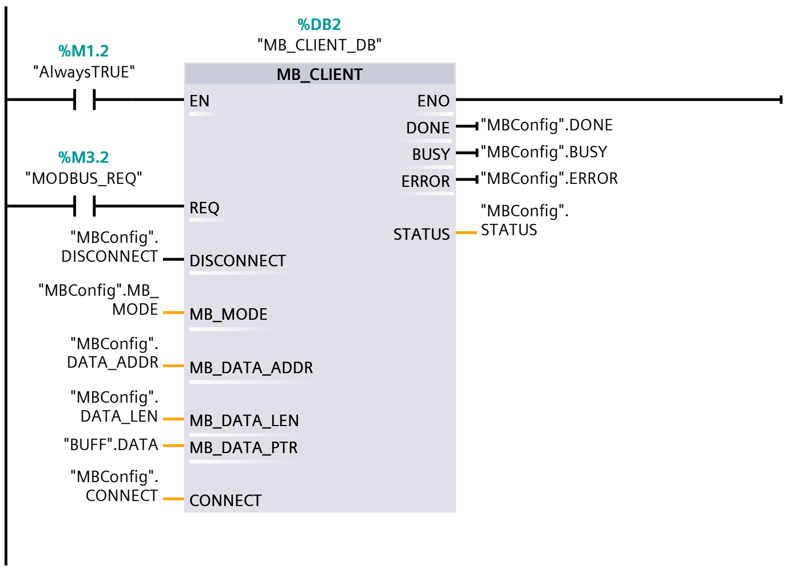
### Network 1:

MODBUS REQ CLOCK CYCLE LOGIC



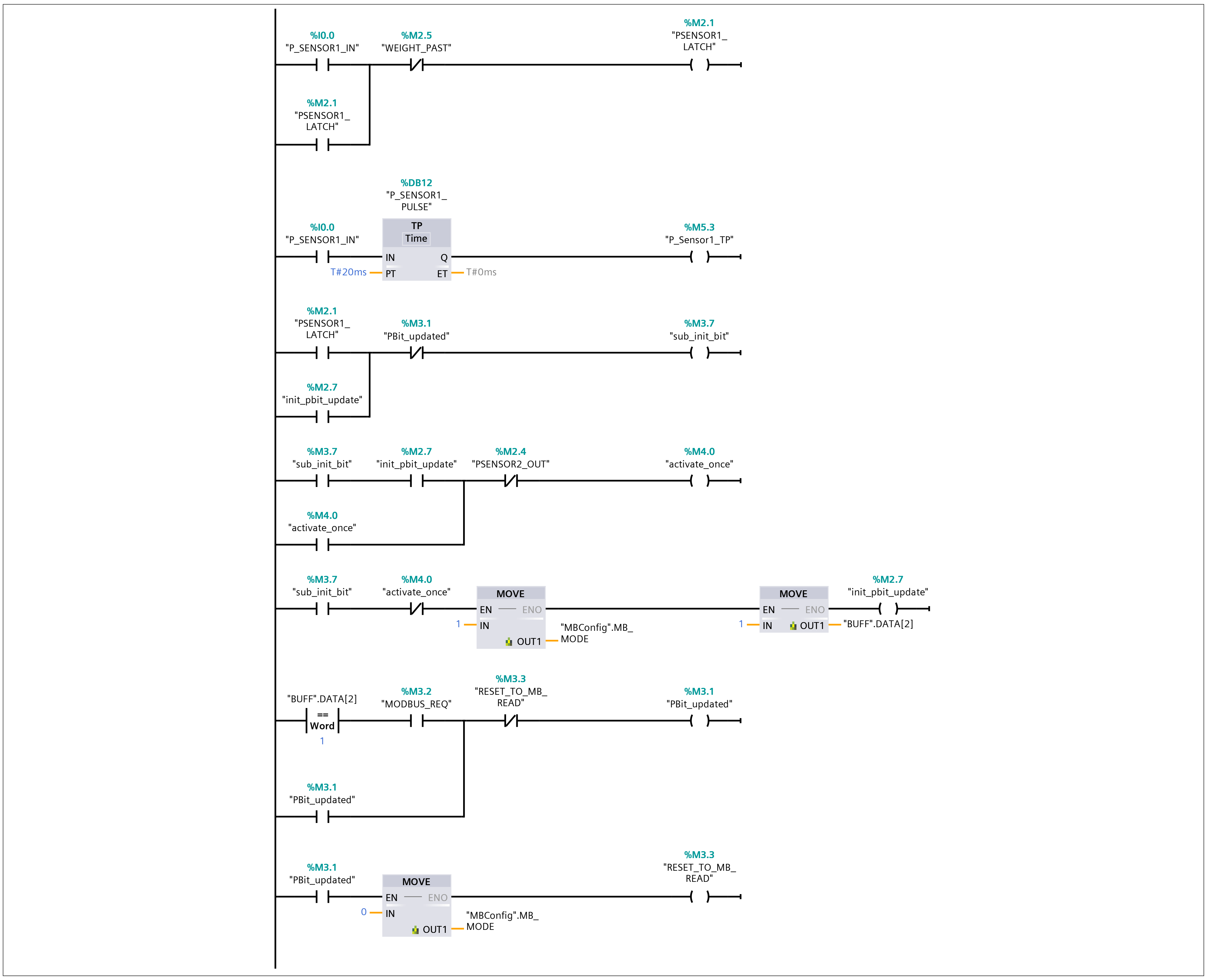

### Network 2:

MODBUS TCP CLIENT SET UP - REFER MBConfig - modbus config pdf for data set up




### Network 3:

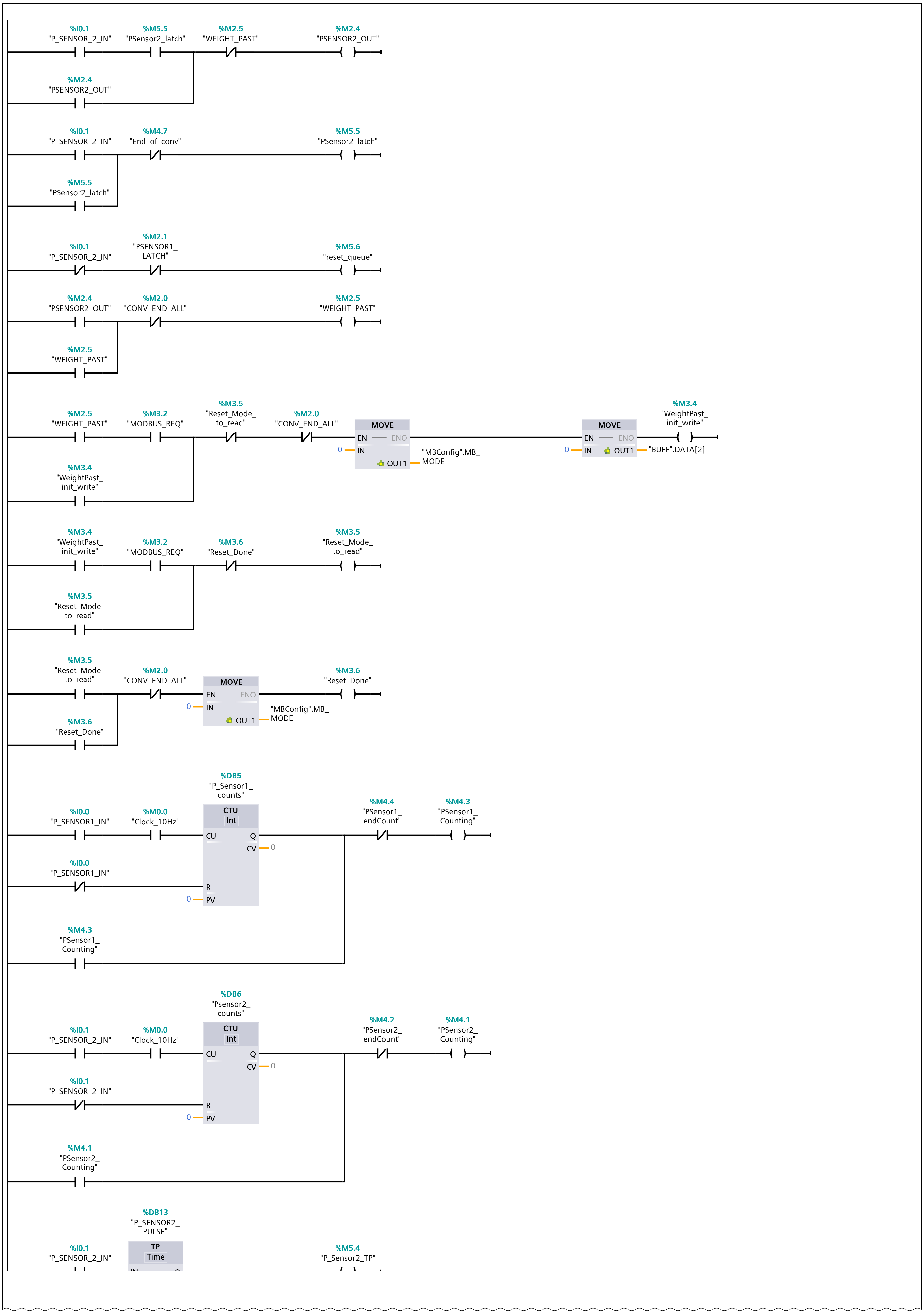
PHOTO ELECTRIC SENSOR 1 LOGIC - LATCH and MODBUS TCP WRITE




## Network 4:

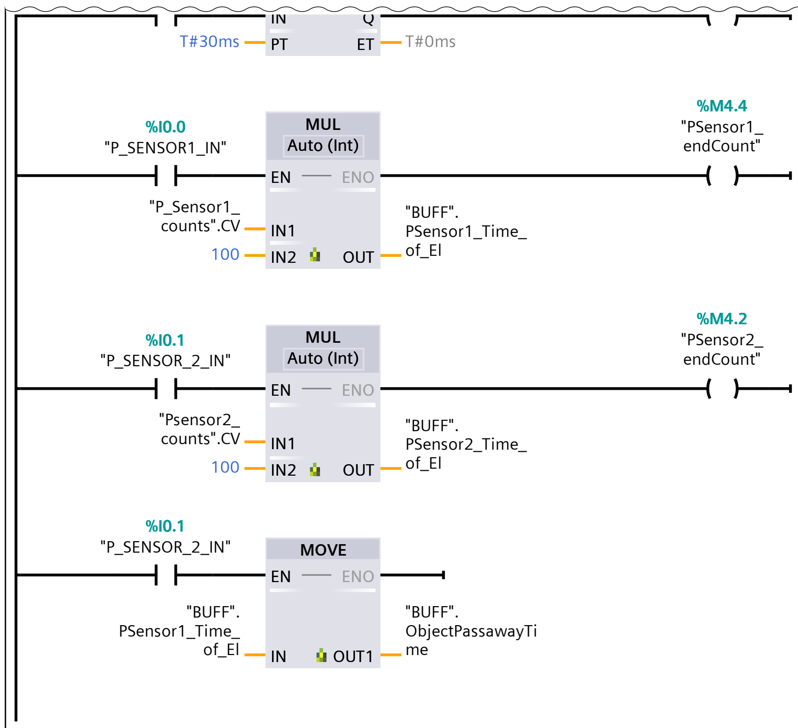
PHOTO ELECTRIC SENSOR 2 LOGIC SETUP, and NECESSARY COUNTER INTEGRATIONS WITH PHOTOELECTRIC SENSOR 1 to help in calculating speed and span of passing elements

**Network 4: (1.1 / 2.1)**



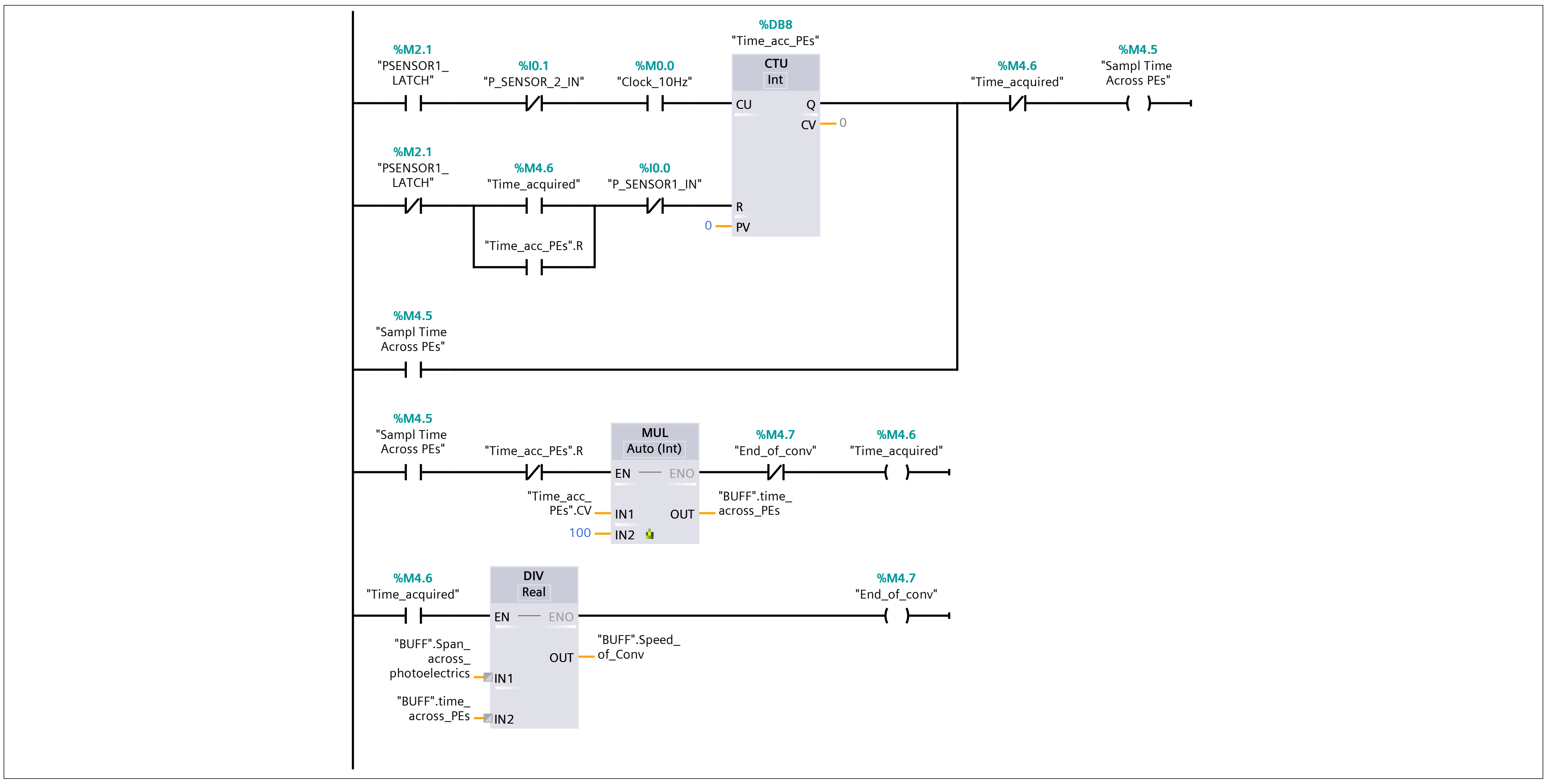
**Network 4: (2.1 / 2.1)**

1.1 ( Page1 - 3)

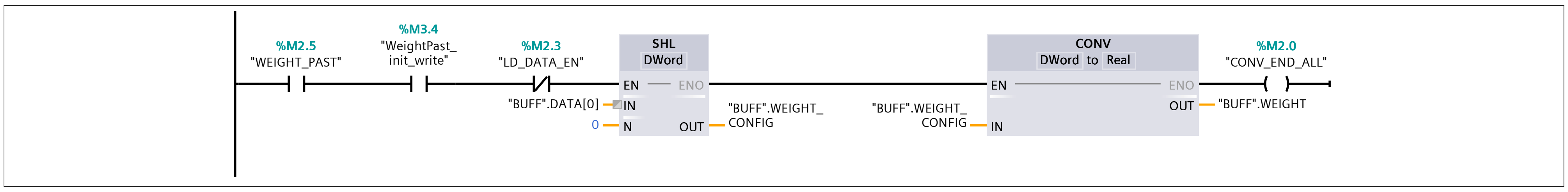


## Network 5: ACQUIRE SPEED OF BELT

SPEED OF BELT LOGIC AND ACQUISITION - TYPICALLY YOU WOULD USE THIS TO DETERMINE WHETHER TO INCREASE OR DECREASE SPEED USING A VFD

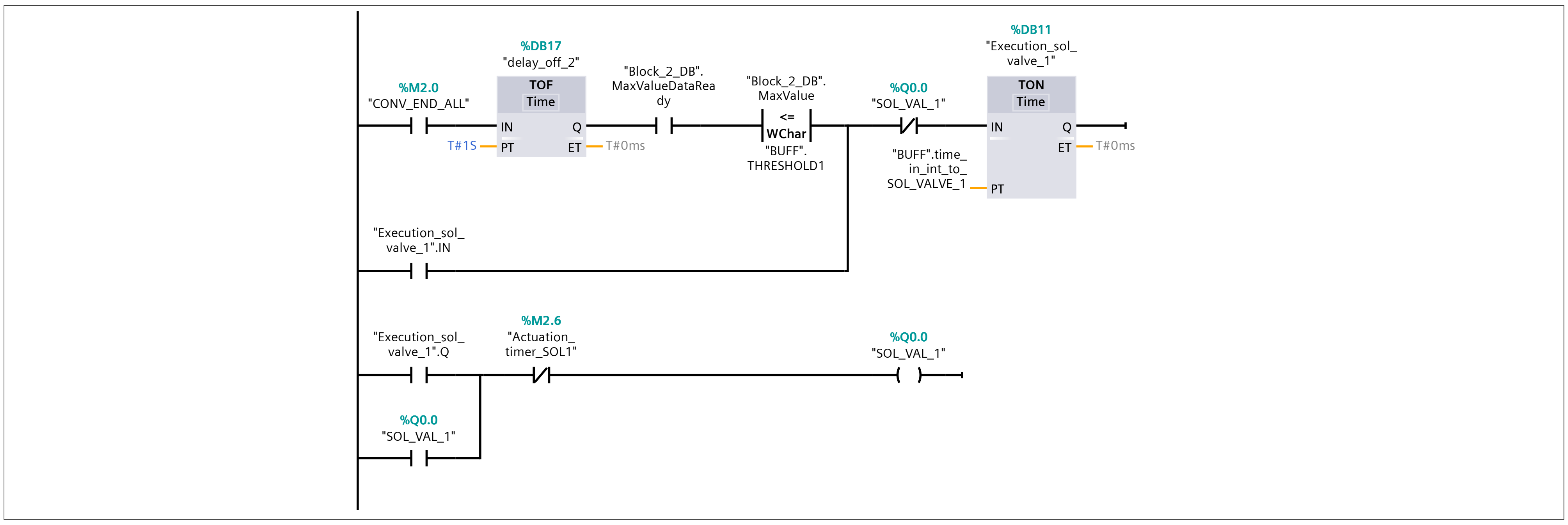



## Network 6:



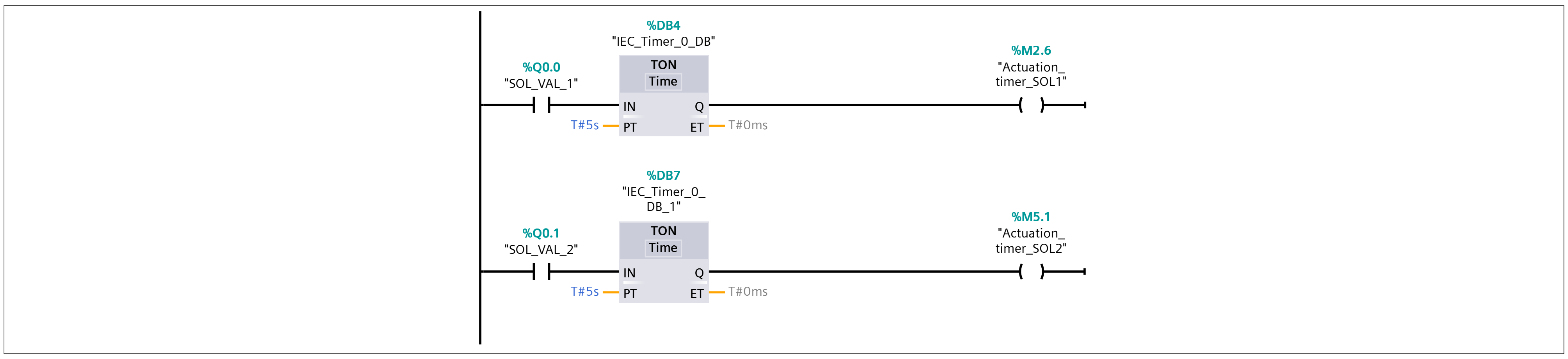

## Network 7:

Rung 1: set delayed execution based on the speed of the belt and the location of the cylinder
Rung 2: Actuate solenoid valve 2 upto set number of seconds check buffer pdf to check set variables



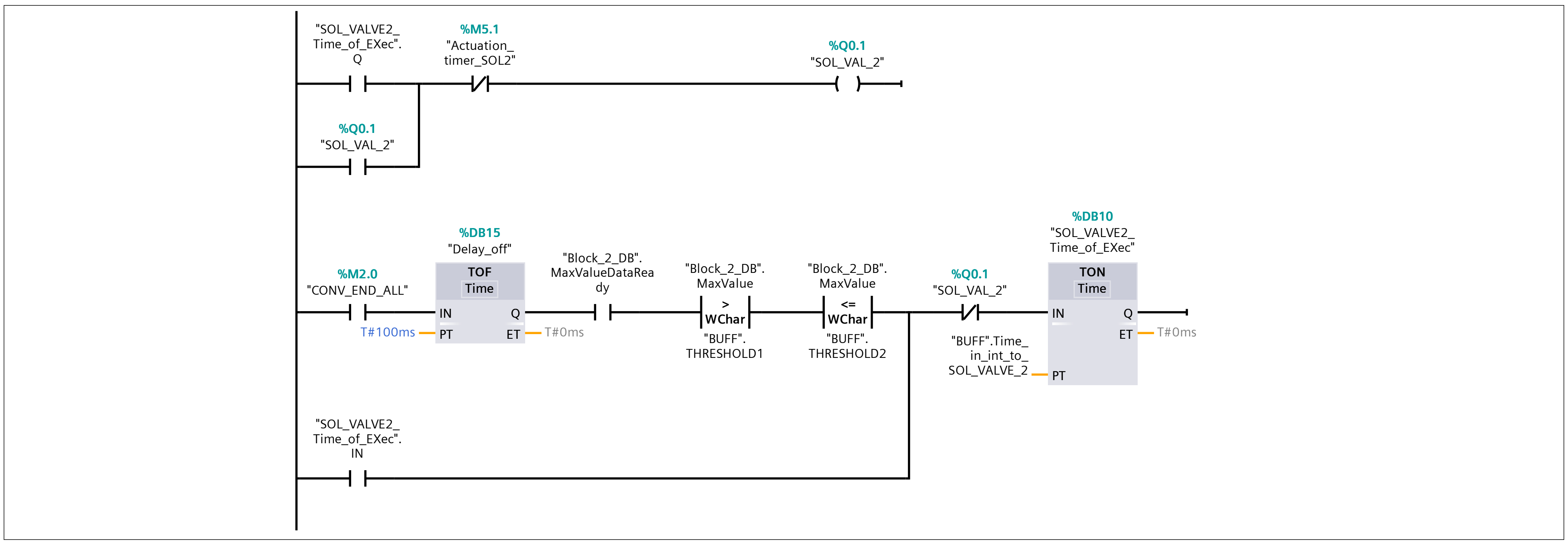

## Network 8:

Timer logic for solenoid valves, typically determine how long the solenoid valves stay on



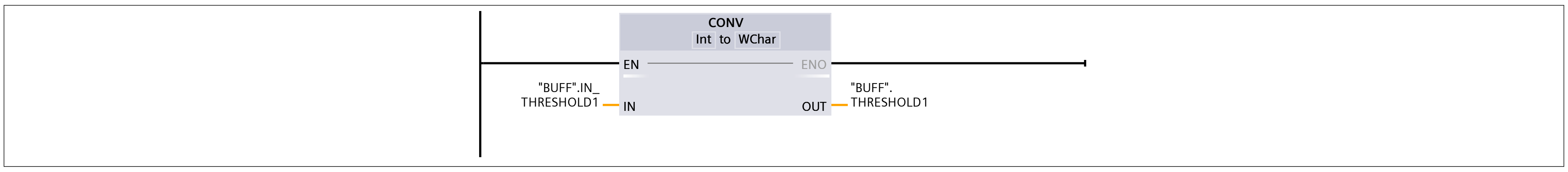

## Network 9:

actuation logic of solenoid valves - refer to network 8 for further details

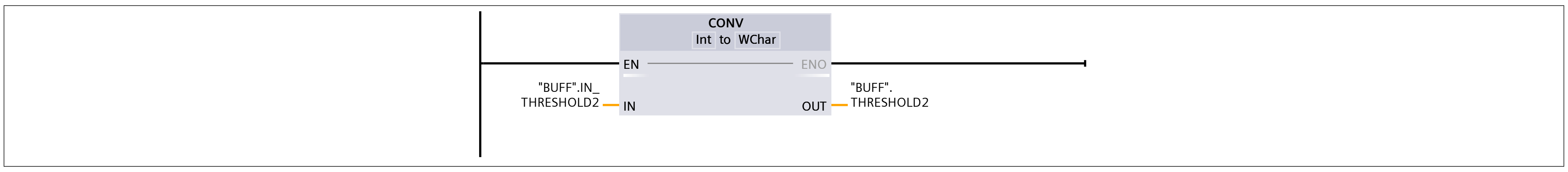



## Network 10:

Conversion threshold 1 from integer to word format

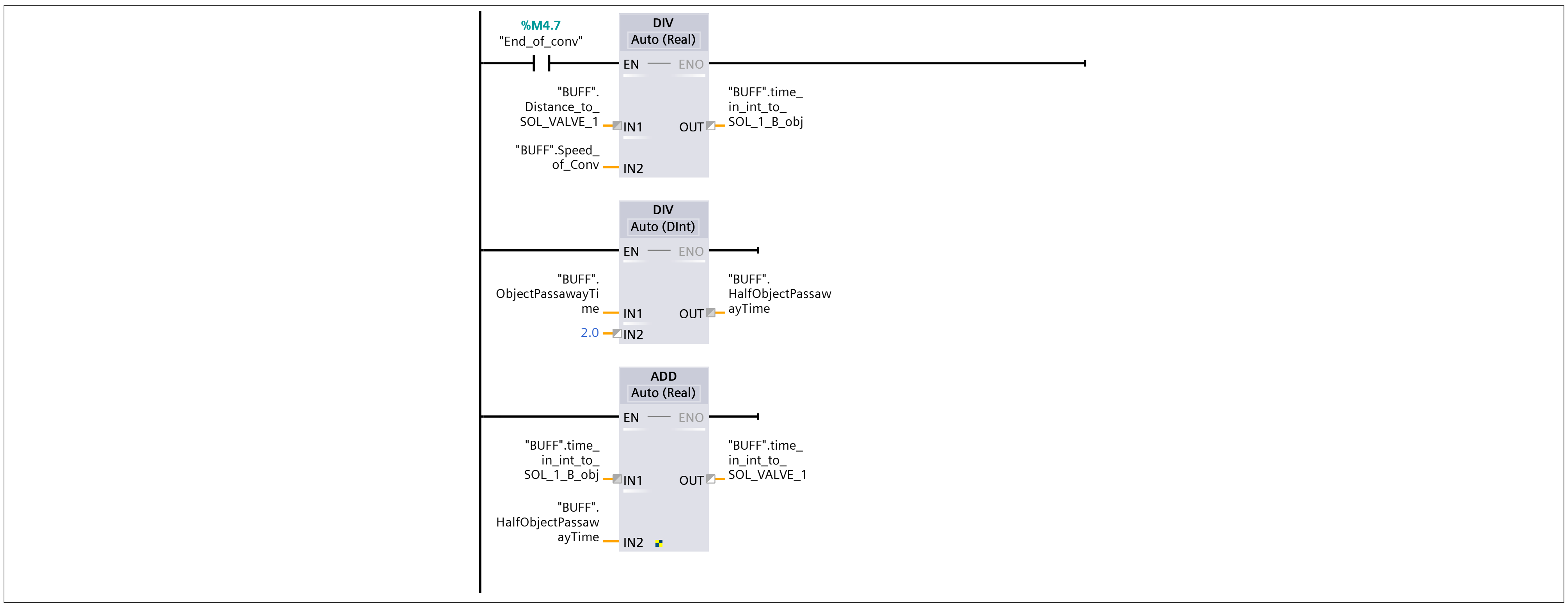



## Network 11:



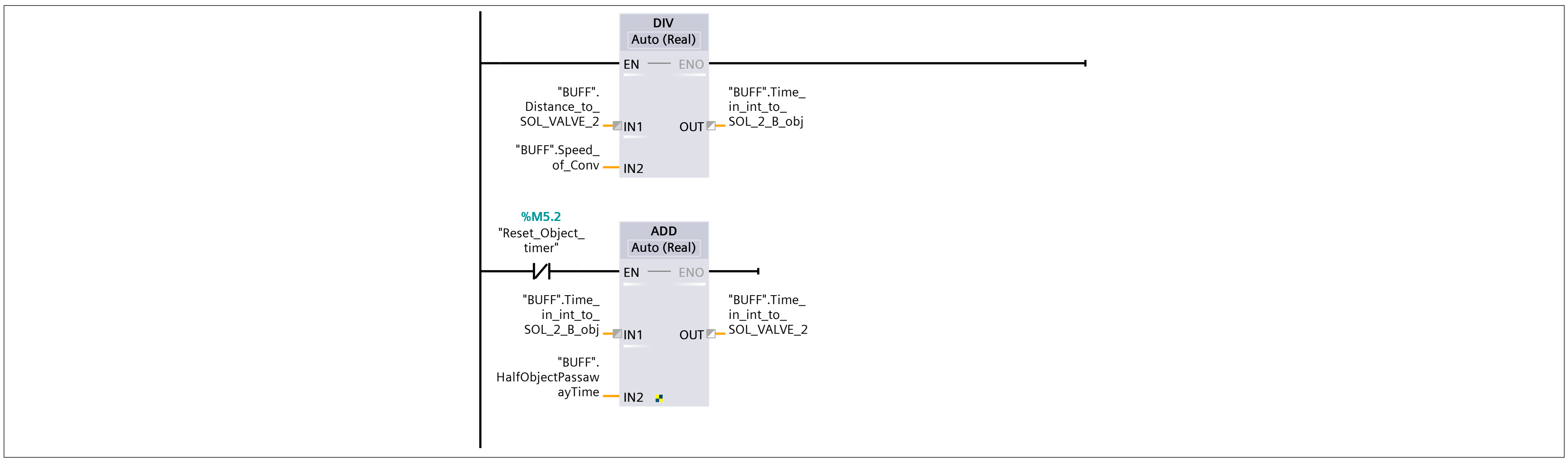

## Network 12:

Division of distance to solenoid valve 1 and speed to calculate time of delay to solenoid valve 1 location. Typical distances to solenoid valves are set at the buffer DB refer -> buffer pdf



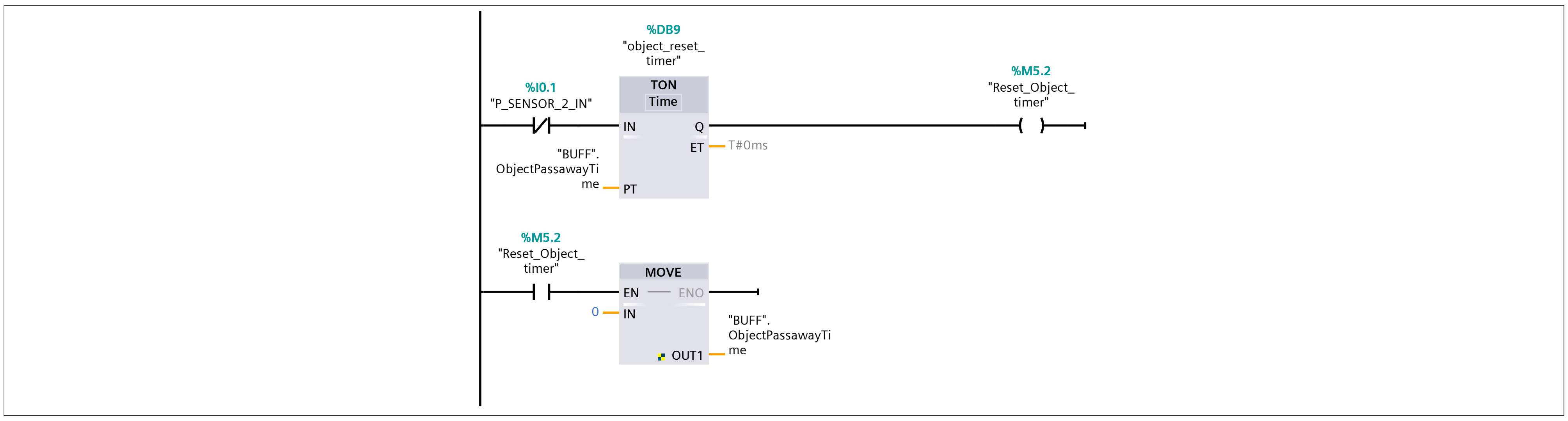

## Network 13:

Calculation of delay time to solenoid valve 2 based on the distance input( Buffer pdf) and speed of the belt



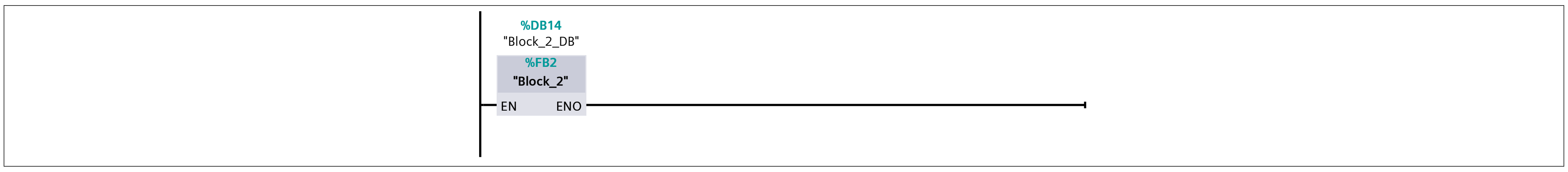

## Network 14:

Handle object timer resets after exit of psensor 2. However the variables are already locked in place by the previous network calculations hence are unaffected by the reset



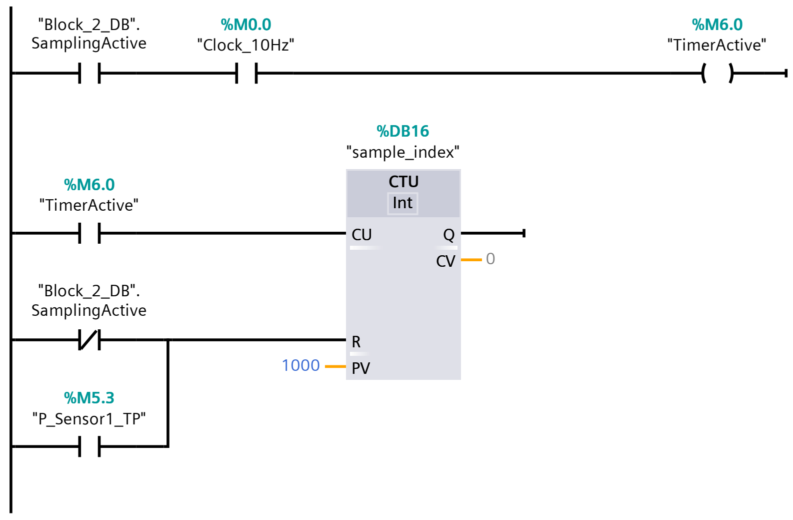

## Network 15:

Ensure that Block_2 [FB2] is active and Structured Context Language Function Block which will be shown in the PDFs - Block_2 uses values from Block_2_DB [DB14]




## Network 16:

Count the sample index to be used in the structured context logic file Block_2 [FB2] refer to PDF


"Block_2_DB".
SamplingActive
%M0.0
"Clock_10Hz"
%M6.0
"TimerActive"
%DB16
"sample_index"
CTU
Int
%M6.0
"TimerActive"
CU
Q
CV
0
"Block_2_DB".
SamplingActive
R
1000
PV
%M5.3
"P_Sensor1_TP"